

EDUCATION

Master of Statistics <i>Graduated with Distinction (75%)</i> Selected Courses: Quantitative and Computational Finance, Financial Econometrics, Financial Risk Management, Causal Inference, Pattern Recognition, Bayesian Inference, Optimization Techniques, Stochastic Processes, Measure Theory. Master's Thesis: Statistical Arbitrage in Electricity Markets Bachelor of Statistics (Hons.) <i>Graduated with Distinction (76%)</i> Selected Courses: Stochastic Processes, Probability Theory, Parametric and Non-Parametric Inference, Statistics Comprehensive, Special Topics in Algorithms, Random Graphs, Operations Research, Numerical Analysis, Linear Algebra	Indian Statistical Institute, Kolkata July 2024 – June 2026 (Advisor: Prof. Diganta Mukherjee) Indian Statistical Institute, Kolkata Sept. 2021 – May 2024
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EXPERIENCE

Goldman Sachs <i>Worked as a Quant Intern, in Risk Economics Strats</i> - Revamped Ongoing Monitoring framework for 1700+ Scenario Design models across Rates, FX, Equity variables. - Designed statistical tests and assessed materiality of breaches through automated filter and exception repository. - Implemented automated documentation of breaches with thresholds, reduces significant manual intervention every cycle. - Built Impact Analysis tools to estimate dollar impact of model changes and performance breaches from firm's sensitivities. BioFin Capital <i>Worked as a Quant Research Intern</i> - Built predictive models (XGBoost, Random Forest) in context of intraday trade signal generation. - Used rule based trades from the signals to obtain 5% market excess return netting transaction costs and slippages. - Developed state of the art models for alpha generation from curated event data across massive set of tickers. Games24x7 <i>Worked as a Research Intern, in Data Science and AI team</i> - Developed and implemented combinatorial metrics for a given hand in Rummy and heuristic driven strategies using them. - Designed 2-player and multiplayer versions of Elo rating for Rummy for efficient matching and skill benchmarking. - Established statistically significant skill element in Rummy, in empirical as well as large scale simulation studies.	May '25 – July '25 Feb '26 - Ongoing Oct '24 - Apr '25
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PROJECTS

Quantile Regression Based Graphical Tail-Dependence in a Panel of Financial Firms - Developed a tail-risk contagion framework using $\tau = 0.1$ quantile regression to estimate conditional VaR spillovers across financial institutions, capturing non-linear dependencies missed by correlation/OLS models. - Implemented a coordinate-descent optimizer for check-loss minimization and validated results against existing packages. - Constructed a tail-dependency graph identifying shock transmitters and stabilizing nodes in the insurance-reinsurance sector. Robust PCA and Background Foreground separation - Explored Robust PCA and Stoch-PCA techniques in context of matrix separation problem. - Studied Convex Envelop to transform NP-hard, non-convex matrix separation problem into tractable solution. - Implemented mentioned algorithms and Online RPCA for real-time separation in dynamic, slowly-moving backgrounds.	Sept. 2025 Guide: Prof. Sumanta Basu April 2025 Guide: Dr. M. Mukherjee
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AWARDS AND ACHIEVEMENTS

Cleared ISI Entrance, JEE Advanced, JEE Mains, WBJEE securing 99.5% + percentile in each	2021
Received Late Ranu Sen Memorial Award	2021
Secured top 1% rank in PRMO, RMO, NSEA	2017,2019,2021
Secured First rank from District in NMMSE	2016-17
Received Scholarship ranking Second in Mathemagic	2018
Won several awards from various mathematics competitions (inc. IFOMG, SumIt, MathGeek, STEMS etc)	

TECHNICAL SKILLS

Languages: Python, R, C, $\LaTeX$, SQL, HTML
Software/Libraries: Git, Numpy, Pandas, Scikit-Learn, TensorFlow, Keras, Django
Statistics and AI: Regression and Diagnostics, Econometrics, Time Series, Classification, Reinforcement Learning

EXTRACURRICULAR ACTIVITIES

Held Joint Secretary position for Integration 2024, conducted MathGeek, MTRP as part of Integration 2023.
Taught 200+ students for mathematics olympiads and entrance examinations.
Enjoy playing chess, various card and board games (e.g. poker, bridge, 29, catan etc), swimming, table tennis.